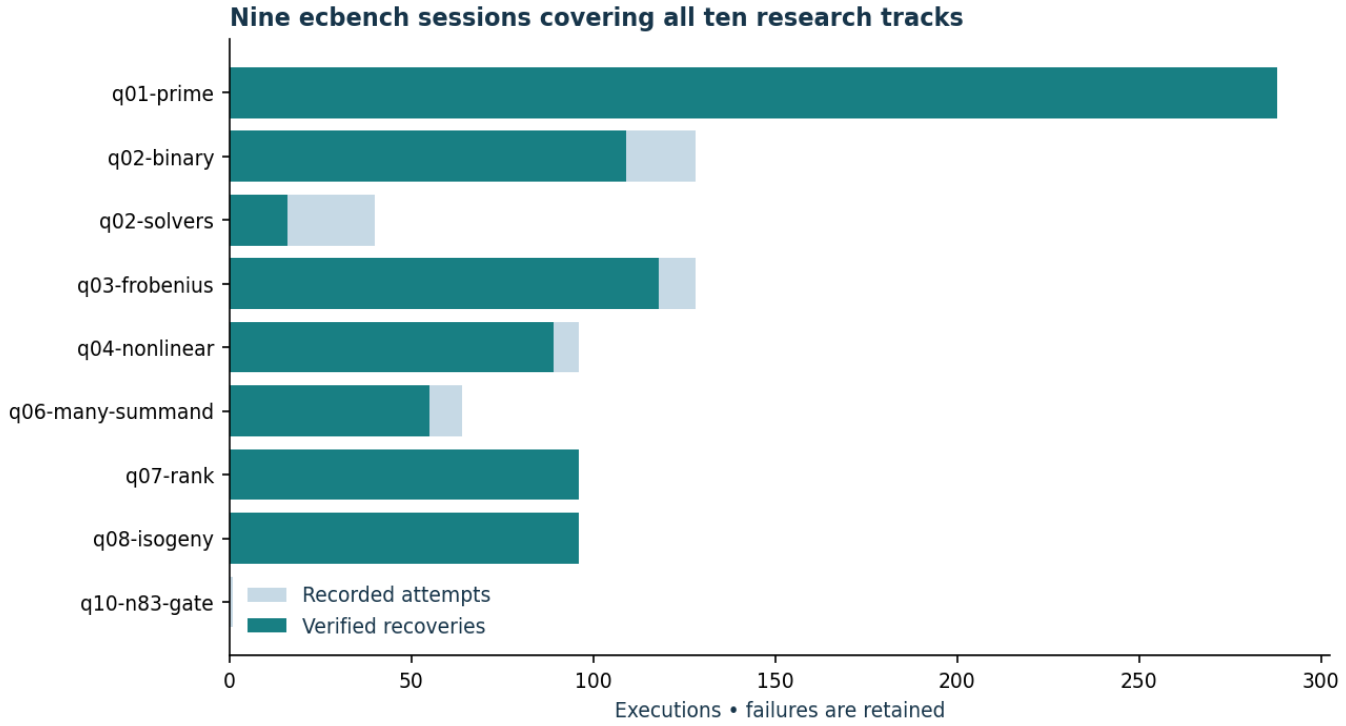
All ten index-calculus questions: ecbench evidence

Prime fields • ordinary binary fields • binary Koblitz curves

937 attempts; 867 verified recoveries; 69 exhausted searches; 1 N83 applicability error.

Nine standard audits passed. 867 verified runs and 70 failure outcomes reproduced. All ten original mathematical questions remain open.

The native harness is aburan28/crypto. Source c716e462934128c75e99dd458748e08623a2708c; binary cc5bba39d33465ddc1b1d31af8ac25c6680d41193fc76012d9a1019b6046f02d; host ECBENV2h687984d4e73d, virtualized AMD EPYC 9V45, L0.



Counts and failures only. No full-cost speedup, asymptotic advance, independent host, admitted wall comparison, GPU or quantum run is claimed. A quotient of partial counts does not bound full cost. The rho controls use the same public points and algorithm seeds; setup is charged cold.

Scope and sources

The ranking and original resolutions come from the source-linked literature/questions.json and literature/REPORT.md, reviewed 2026-10-14. Recent sources include Caminata–Gorla (2023), Bernstein–Lange (2024), the September 2026 Hamming-ideal and annealing papers, plus the 2020/2021 subfield and quasi-subfield work. Gamma-weight point bases are distinct from syndrome-decoding Hamming ideals.

1. Can prime-field curves admit a sparse factor base with genuinely cheap decomposition?

Status: partial. A scalable prime-field base with executable cheap membership/decomposition and a fully priced held-out crossover.

Repo/paper basis: R2; R3; R16; P10; P2

2. What is the actual solving degree of binary summation systems as the parameters grow?

Status: partial. Growing-dimension solving-degree bounds; correct chained-system shape metadata and held-out satisfiable/infeasible queries.

Repo/paper basis: R4; P1

3. Can Frobenius orbit decomposition retain a compact algebraic witness?

Status: partial. Compact algebraic witnesses for relative Frobenius phases, including membership, splitting and lift costs.

Repo/paper basis: R5; R6; P6

4. Which Frobenius-stable nonlinear factor bases keep both yield and decomposition cost favorable?

Status: partial. Scalable nonlinear algebraic membership/solving; gamma weight is not a normal-basis Hamming ideal.

Repo/paper basis: R7; R6; P3; P6

5. Can explicit quasi-subfield constructions become useful at prime extension degrees?

Status: partial. A useful prime-degree parameter family, subgroup yield law and complete measured IC integration.

Repo/paper basis: R8; R4; P8; P9

6. Can multi-summand formulations avoid paying back their degree reduction during lifting?

Status: partial. Growing arity with full splitting/lifting/matrix costs; literal ten-summand toy results do not transfer to N83.

Repo/paper basis: R4; R10; R9; P7; P1

7. How can a collector buy independent rank and reliable target descent efficiently?

Status: partial. Rank and descent reliability laws, optimized large-prime policies and fully priced reusable setup/online windows.

Repo/paper basis: R4; R9; R11; P6

8. Can isogeny and Hecke mechanisms overcome their rank and transport barriers?

Status: partial. Explicit paid isogeny transport and a Hecke-rank theorem or certified counterexample in its stated model.

Repo/paper basis: R3; R12; R13; R18; P2

9. Can a higher-genus cover make the elliptic component cheaper to solve?

Status: partial. Independent exact source-order proofs, a general Jacobian norm and a cheaper higher-genus relation/descent solver.

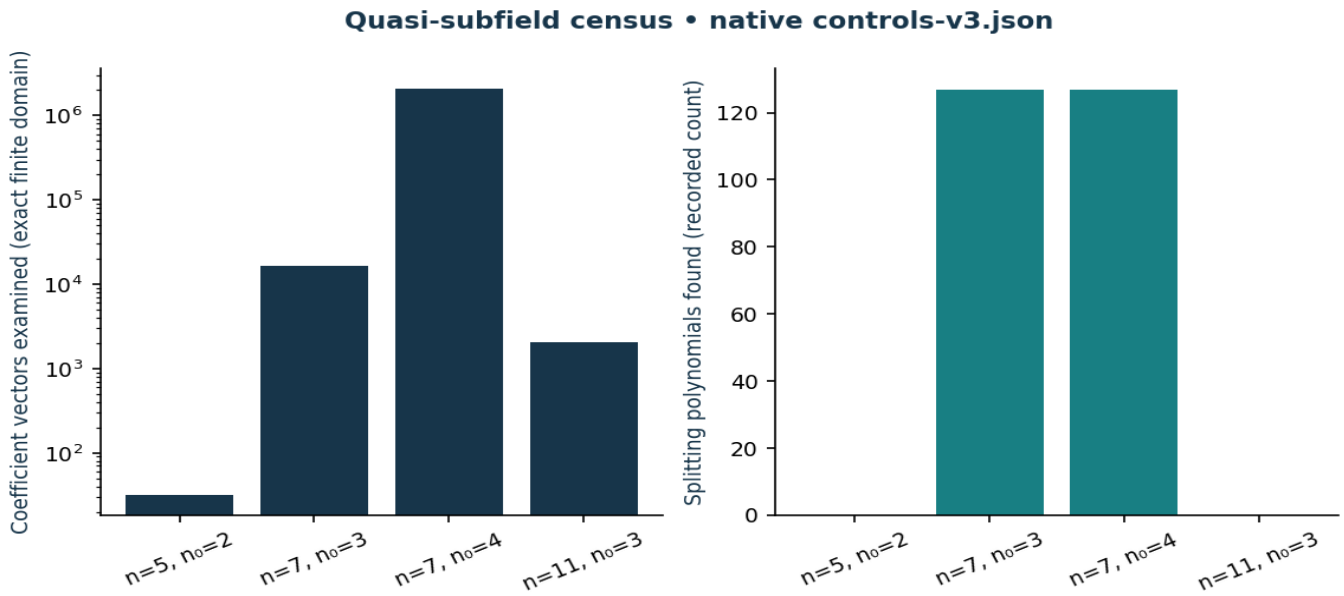
Repo/paper basis: R14; R15; P11

10. Where is the verified complete-attack crossover with tuned Pollard rho?

Status: partial. Wide IC at N83, N131 transfer justification, isolated A/A, independent host, complete pricing, held-out scaling and matched GPU/annealing hardware.

Repo/paper basis: R11; R16; P4; P5; P2

Structural observations and limits



Independent root checks verify finite witnesses, not a general family or natural-query complexity law. Actual subgroup support is recorded per witness.

The actual chained system has intermediate field variables beyond the factor-base coordinates. First fall degree, Boolean roots and full point-decomposition success are separate observations.

Degree-83 actual genus-3 image classes passed annihilation and sampled homomorphism/push-pull checks. The prime-field genus-2 certificate is valid, while its arithmetic and transport remain unsupported. Generic toy Jacobian recovery supplies a functional check only.

N83 IC was attempted and returned the missing-wide-path error. Degree 31 has $\mathrm{ord}_{31}(2)=5$ and does not replace N83/N131 module structure. E1's smaller degree-83 subgroup does not replace E0's required subgroup.

The exact ten-summand toy experiment ran without downscaling. The raw sessions preserve every failed attempt and native resource vector. Primary online windows are recorded, but their L0 wall clocks are inadmissible.

Reproduce and review

Protocols, sealed records, standard audits, failure replay inputs/outputs, locked dependencies, source hashes, native tools and per-run reports are beside this report. No new mathematical theorem or independent source-order certificate is claimed.

Repository evidence directory